MACHINE LEARNING · EXPLAINABILITY · CASE STUDY

What Drives Medical Insurance Costs?

An end-to-end data science project — from raw data to an explainable, actionable pricing strategy.

Subhransu Dhar, PhD

3.8x	R2 0.85	$41,558	~$15.3M
smoker cost gap	cross-validated	priciest segment	annual opportunity

THE CHALLENGE

The Problem

Insurers live or die by how well they predict cost.

- Health insurers must predict what a customer will cost —
- to price policies fairly, flag high-risk members, and target
- wellness spend.

- Mispricing cuts both ways: too high loses customers to
- competitors; too low loses money on claims.

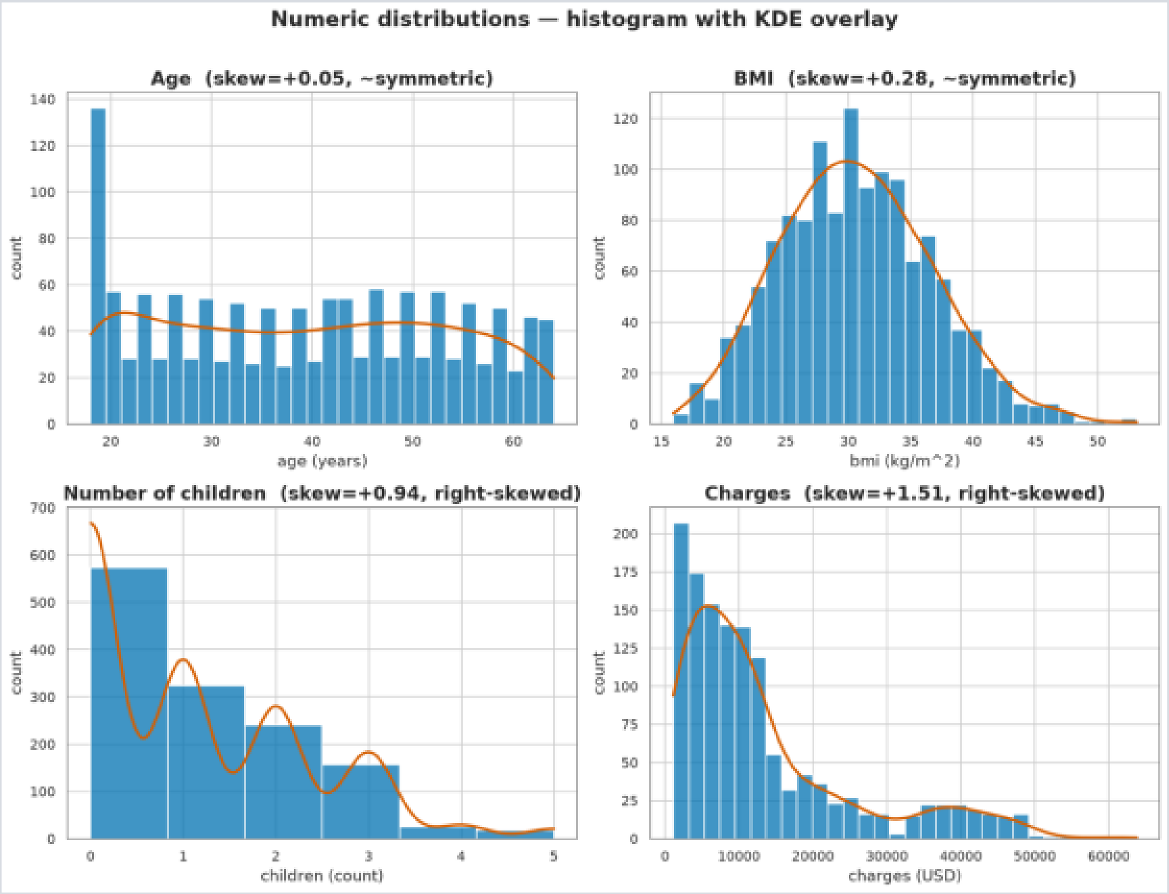
1. **What drives medical cost?**
2. **Can we predict it accurately and transparently?**
3. **What should the business do differently?**

CHAPTER 1

The Data

- 1,337 policyholders after cleaning
- 7 features: age, sex, BMI, children,
- smoker, region -> charges (target)
- 0 missing values; 1 duplicate removed
- ~20.5% smokers; balanced sex & region

The target is heavily right-skewed —
a dense low-cost cluster and a long
high-cost tail (almost all smokers).



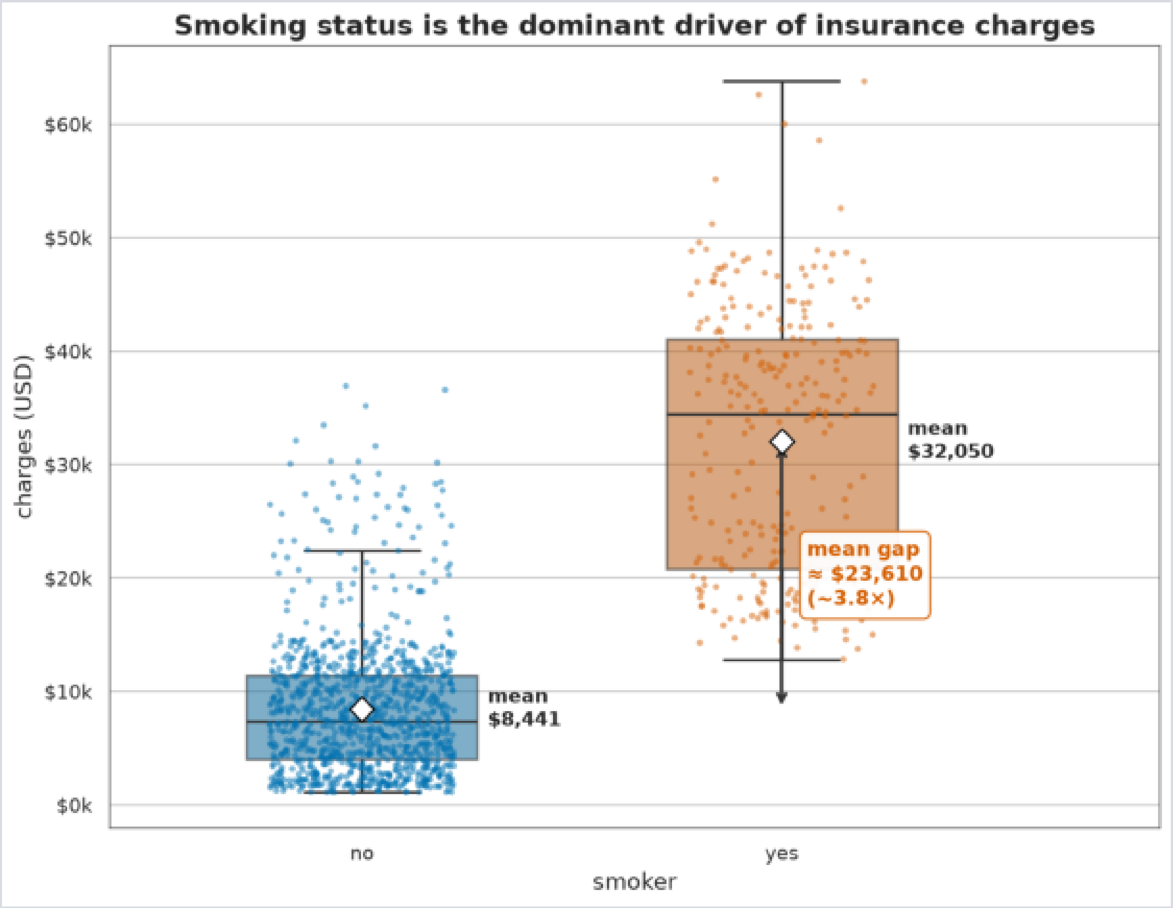
FINDING 1

Smoking Dominates

3.8x

smoker vs non-smoker cost

- $32,050 avg for smokers vs $8,441 for
- non-smokers — the strongest signal
- Smokers are 20.5% of members but
- ~50% of all claims cost
- The two populations barely overlap



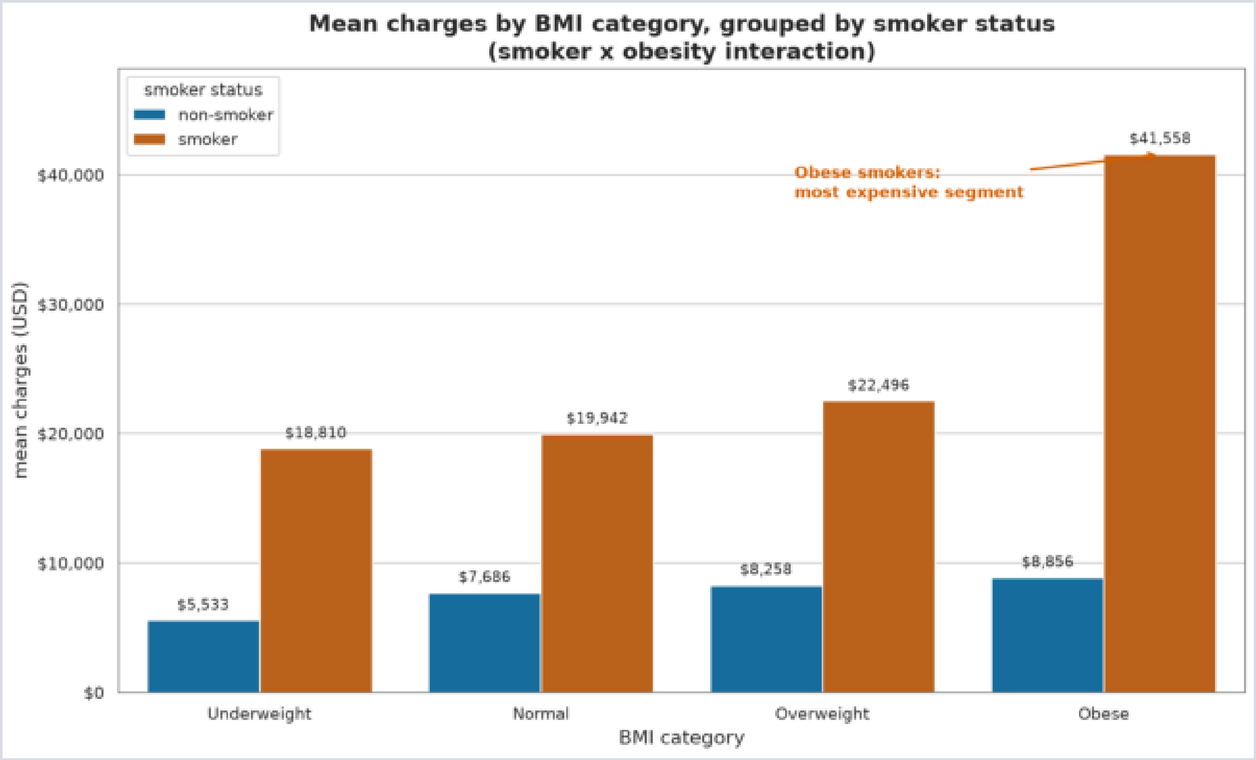
FINDING 2 · THE KEY INSIGHT

Smoking x Obesity Compounds

$41,558

avg cost of an obese smoker

- Obese smoker $41,558 vs non-obese
- smoker $21,363
- Obese NON-smoker: just $8,856
- BMI barely matters unless you smoke —
- the risk multiplies, it doesn't add



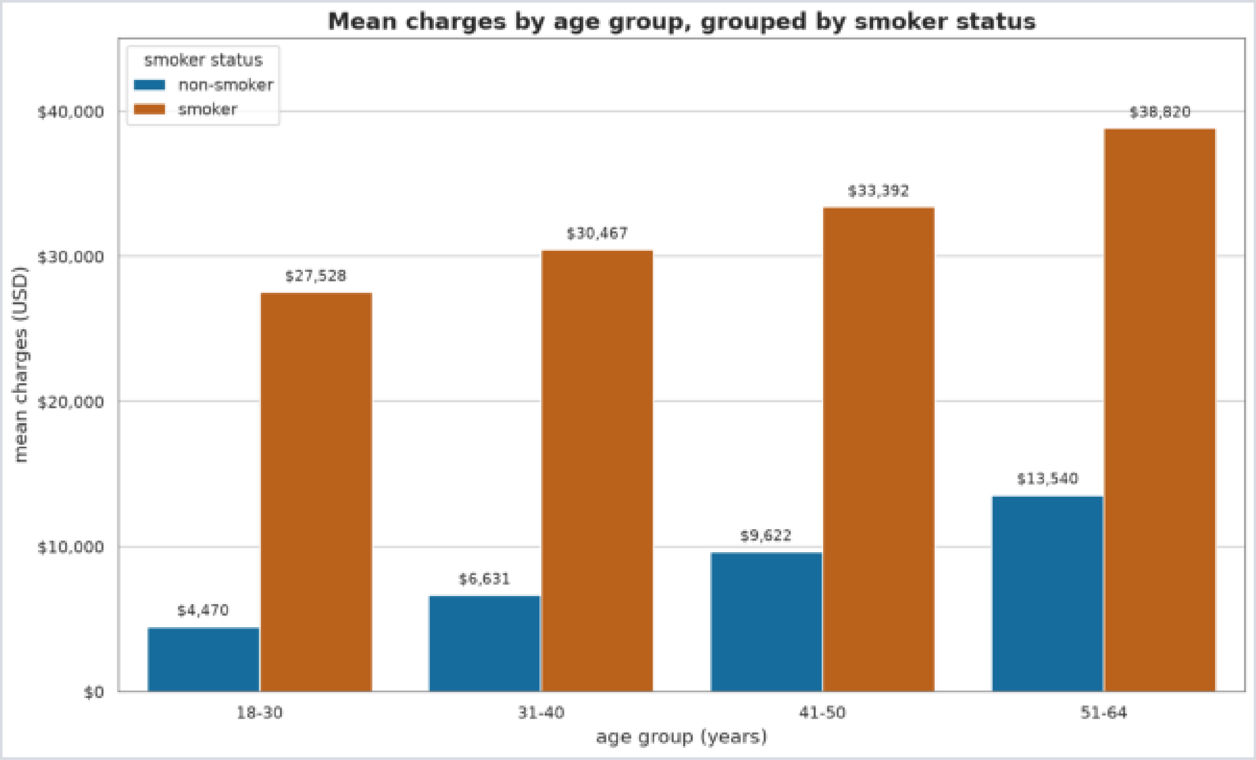
FINDING 3 (+ WHAT ISN'T A DRIVER)

Age is Secondary

~$267

added cost per year of age

- Age adds ~$2,700 per decade, for
- smokers and non-smokers alike
- The smoker gap stays wide at every age
- Region: NOT a reliable driver
- (fails variance-robust tests)
- Sex: NOT a driver (Cohen's d ~ 0.12)



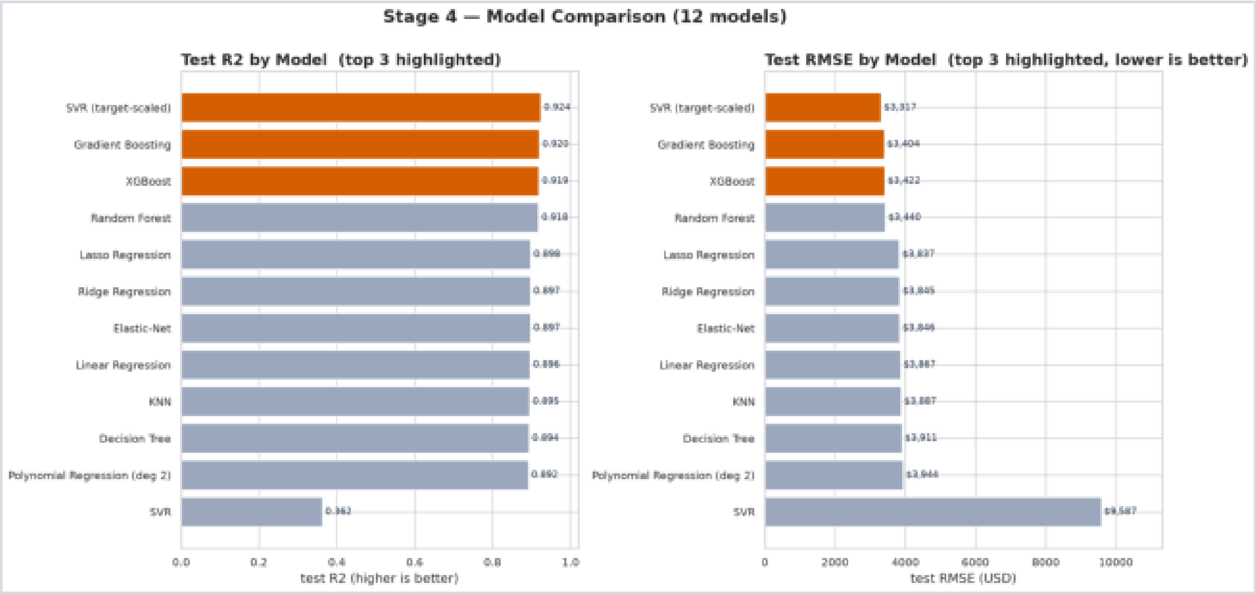
CHAPTER 3

The Model — 17 Configurations

R2 0.85

cross-validated (honest)

- Linear, tree ensembles, SVR, KNN,
- voting & stacking — all grid-tuned
- Top ~6 models are a statistical tie
- Chose Gradient Boosting: accurate,
- simple, natively explainable
- RMSE ~$3,400; quoted CV 0.85 not the
- flattering single-split 0.92



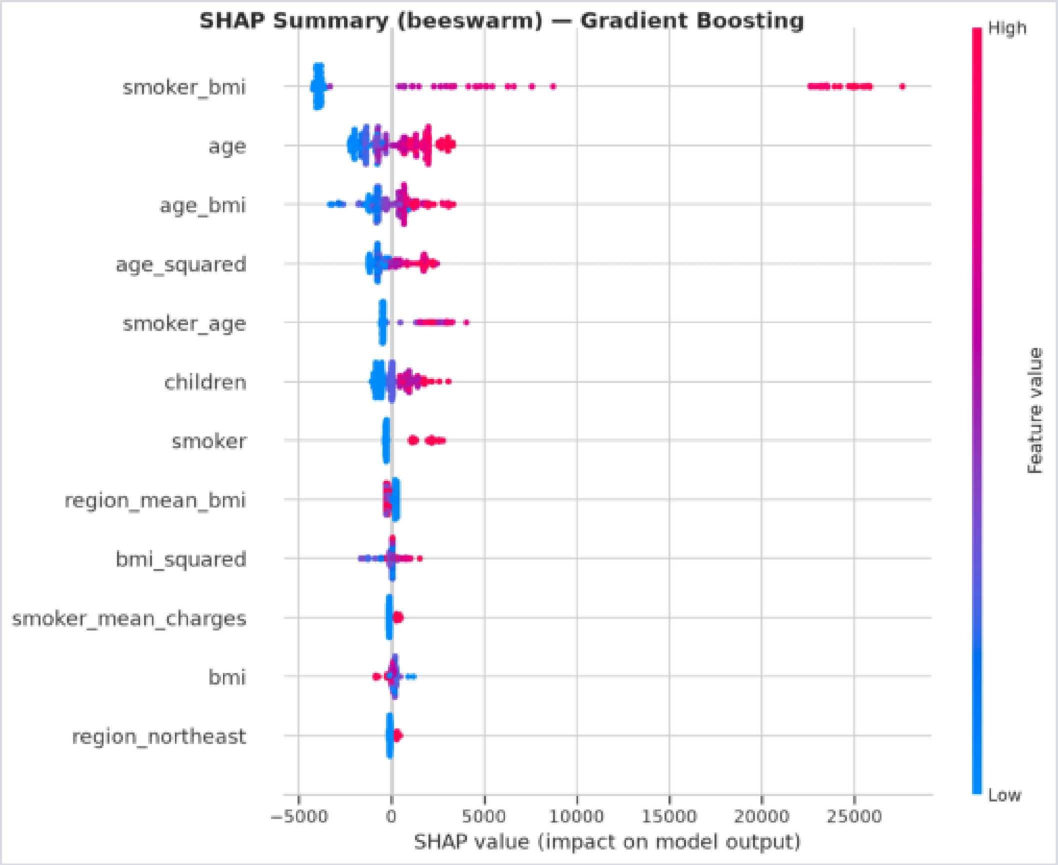
CHAPTER 4

Why the Model Predicts What It Does

62%

of predictive weight = smoking

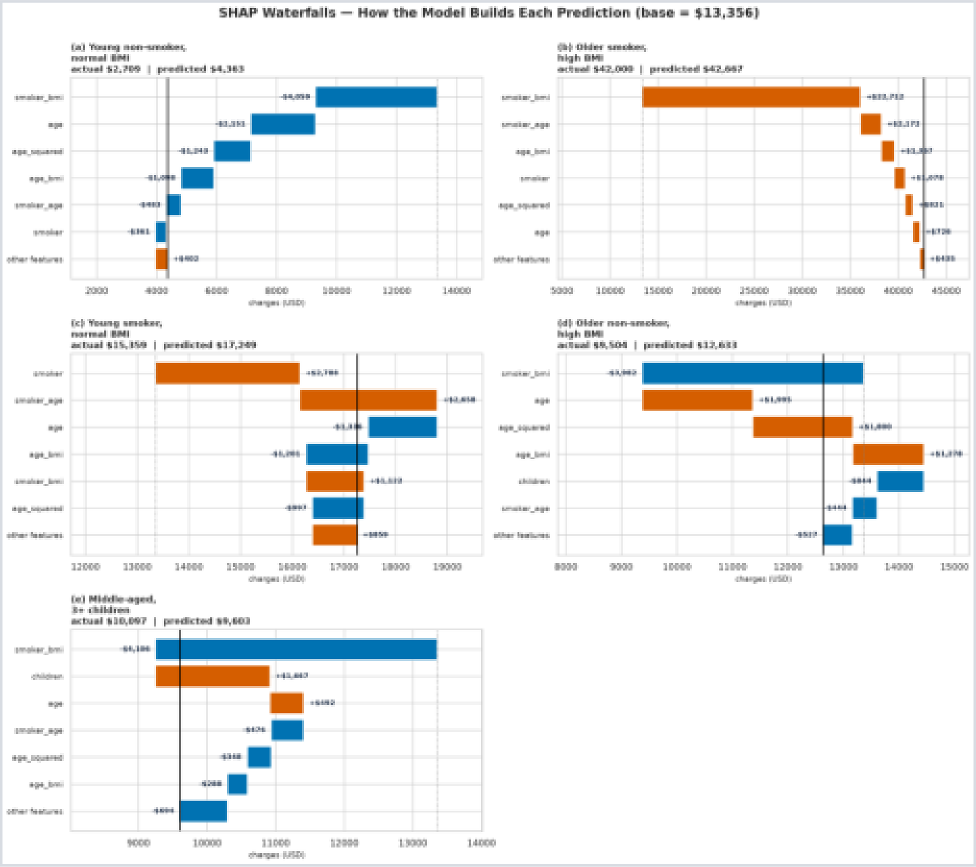
- SHAP assigns each feature a signed
- dollar contribution per prediction
- smoker x BMI swings a quote by up to
- +$25,000, stepping up at BMI 30
- Region + sex combined: just 3.6%
- Explainable AI, not a black box



CHAPTER 4

Every Prediction, Explained

Per-quote transparency: each estimate is built from a $13,356 baseline, feature by feature — a $4,363 young non-smoker vs a $42,667 obese smoker.



Recommendations

1. **Reprice on a smoker x BMI tier**

   Not a flat surcharge — today under-prices obese smokers by ~$20,000 each.

2. **Fund smoking cessation + targeted weight mgmt**

   Biggest addressable pool; target obesity at smokers specifically.

3. **Acquire claims & diagnosis data**

   The model's worst misses are high-cost non-smokers — drivers not yet captured.

4. **Deploy transparently with SHAP**

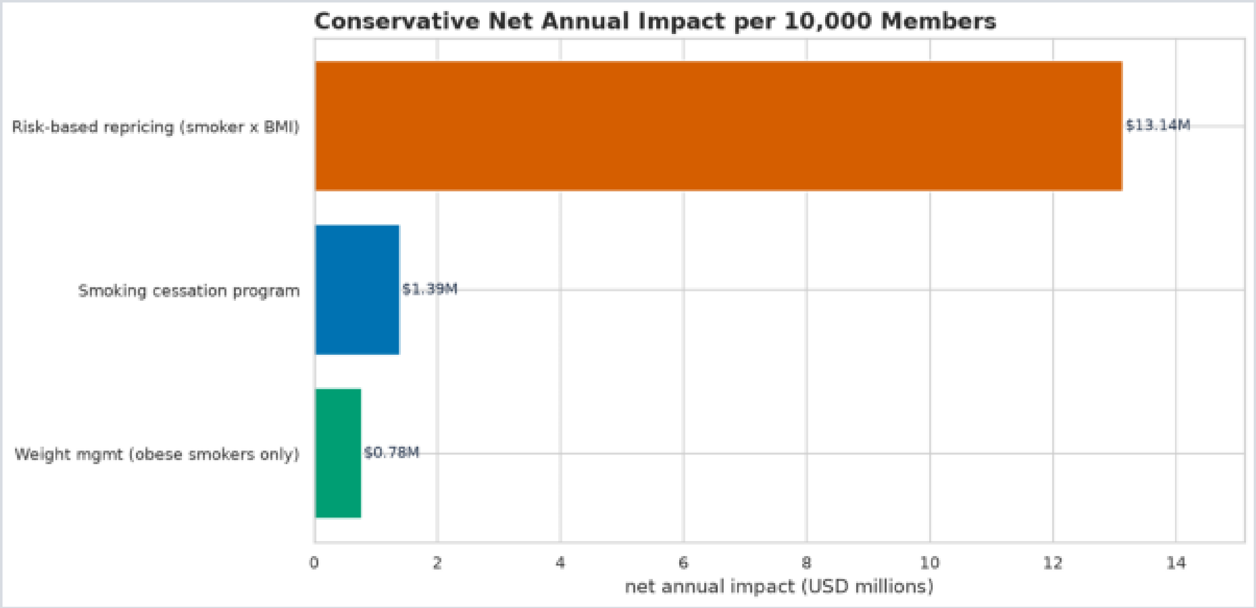
   Per-quote explanations; exclude sex & region as rating factors (fairness).

Financial Impact

~$15.3M

per year / 10,000 members

- Risk-based repricing: ~$13.1M
- (premium adequacy)
- Smoking cessation (10% quit): ~$1.39M
- Targeted weight management: ~$0.78M
- The biggest lever costs almost
- nothing to implement



Thank You

One in five members drives half the cost; smoking-and-obesity is the engine — and now we can price, target, and explain that risk.

Email: subhransu.dhar@gmail.com

Live report: subhransudhar-projects.github.io/medical-insurance-cost-prediction

Code: github.com/subhransudhar-projects/medical-insurance-cost-prediction

Portfolio: subhransudhar-projects.github.io

Subhransu Dhar, PhD · Vienna, Austria